

Assignment-1

Name - Sourav

Reg No - 23FE10CSE00326

Subject - Statistics and Probability (MAS2001)

Section - I

Section-A

Q1 (a) Define random variable

Ans → A Real valued function defined on sample space 'S' taking values on $R (-\infty, \infty)$ is called a Random variable.

(b) If two random variable X and Y are independent, then...

$$E(XY) = E(X)E(Y)$$

$$E(X+Y) = E(X) + E(Y)$$

(c) Write the statement of Chebyshev's inequality.

If X is random variable with $E(X) = \mu$ and $V(X) = \sigma^2$

then $* P(|X - \mu| < k\sigma) = P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$

$$* P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$* P(|X - \mu| \geq x) \leq \frac{\text{var}(X)}{x^2}$$

(k and x are constants)

(d) Define distribution function.

If X is Random variable, then its distribution function is a function $F_X: R \rightarrow [0, 1]$

$$F_X(x) = P(X \leq x) \quad \forall x \in R$$

(e) Is the function defined as follows a density function?

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Condition $\Rightarrow$

$$(i) f(x) \geq 0 \Rightarrow f(x) = e^{-x} \geq 0 \Rightarrow$$

$$(ii) \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_{-\infty}^{\infty} e^{-x} dx$$
$$= \int_{-\infty}^0 0 dx + \int_0^{\infty} e^{-x} dx$$

$$= 0 + 1$$

$$= 1$$

→ Yes! function $f(x)$ follows a density function

(f) A discrete random variable can take finite or countable infinite number of values within its range

(g) The range of distribution function is $[0, 1]$

(h) If X and Y are two random variables such that their expectations exist and $P(X \leq Y) = 1$ then

$$\Rightarrow E(X) \leq E(Y)$$

(i) The height of a person in a country is a random variable of the type...

Ans $\rightarrow$ Continuous Random Variable

Section B:-

Q-2 Two dice are rolled. Let X denote the random variable which counts the total number of points on the upturned faces, construct a table giving the non-zero values of the probability mass function. Also find the distribution function of X .

(sum)
Number of ways to get sum
Total ways

X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$\rightarrow$ Distribution function $\Rightarrow$

$$F_X(X < 2) = 0$$

$$F_X(X < 3) = \frac{1}{36}$$

$$F_X(X < 4) = \frac{3}{36}$$

$$F_X(X < 5) = \frac{6}{36}$$

$$F_X(X < 6) = \frac{10}{36}$$

$$F_X(X < 7) = \frac{15}{36}$$

$$F_X(X < 8) = \frac{21}{36}$$

$$F_X(X < 9) = \frac{26}{36}$$

$$F_X(X < 10) = \frac{30}{36}$$

$$F_X(X < 11) = \frac{33}{36}$$

$$F_X(X < 12) = \frac{35}{36}$$

$$F_X(X < 12) = \frac{36}{36} = 1$$

$$F_x(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{36}, & 2 \leq x < 3 \\ \frac{3}{36}, & 3 \leq x < 4 \\ \frac{6}{36}, & 4 \leq x < 5 \\ \frac{10}{36}, & 5 \leq x < 6 \\ \frac{15}{36}, & 6 \leq x < 7 \\ \frac{21}{36}, & 7 \leq x < 8 \\ \frac{26}{36}, & 8 \leq x < 9 \\ \frac{30}{36}, & 9 \leq x < 10 \\ \frac{32}{36}, & 10 \leq x < 11 \\ \frac{35}{36}, & 11 \leq x < 12 \\ \frac{36}{36} = 1, & x \geq 12 \end{cases}$$

Q.8

A diameter of an electric cable, say X , is assumed to be a continuous random variable with p.d.f. : $f(x) = 6x(1-x)$, $0 \leq x \leq 1$.

(i) Check that $f(x)$ is pdf, and (ii) determine a number b such that

$P(X < b) = P(X > b)$. (iii) Obtain expression for the cdf of X

(iv) Compute $P(X \leq \frac{1}{2} \mid \frac{1}{2} \leq X \leq \frac{2}{3})$

① $\rightarrow f(x) \geq 0$

$$\rightarrow \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} 6x(1-x) dx$$

$$= \int_0^1 (6x - 6x^2) dx$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 6 \left[\frac{1}{2} - \frac{1}{3} \right] = ① =$$

do
 $f(x)$ is pdf

$$\begin{aligned}
 \textcircled{ii} \quad P(X < b) &= \int_0^b f(x) dx \\
 &= 6 \int_0^b (x - x^2) dx \\
 &= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^b \\
 &= 3b^2 - 2b^3 \quad \text{--- (1)}
 \end{aligned}$$

$$\begin{aligned}
 P(X > b) &= 1 - P(X < b) \\
 &= 1 - 3b^2 + 2b^3 \quad \text{--- (2)}
 \end{aligned}$$

$$\because P(X < b) = P(X > b)$$

$$\therefore 3b^2 - 2b^3 = 1 - 3b^2 + 2b^3$$

$$\Rightarrow 6b^2 - 4b^3 - 1 = 0$$

$$\begin{aligned}
 \Rightarrow \quad \cancel{2b^3 - 4b^3} \oplus b &= \frac{1 + \sqrt{3}}{2} = 1.366 \\
 \ominus b &= \frac{1 - \sqrt{3}}{2} = -0.366
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} \Rightarrow \end{aligned}} \right\} \text{Not in Range } [0, 1] \text{ as mentioned}$$

$$* b = \frac{1}{2} = 0.5 \checkmark$$

hence $\boxed{b = \frac{1}{2}}$ why

$\textcircled{iii} \quad \text{CDF} \Rightarrow$

$$F(x) = \int_{-\infty}^x f(x) dx \Rightarrow$$

$$\text{I} \quad x \leq 0$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x 0 dx = 0$$

$$\text{II} \quad 0 < x < 1$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= \int_0^x 6x(1-x) dx$$

$$= \left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_0^x$$

$$= 3x^2 - 2x^3$$

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$$F(x) = 0$$

So Cumulative distribution function =

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 3x^2 - 2x^3, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases} \text{ Ans}$$

(iv)

$$P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right) = \frac{P\left(X \leq \frac{1}{2} \cap \frac{1}{3} \leq X \leq \frac{2}{3}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)}$$

$$= \frac{P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)}$$

$$= \frac{\int_{1/3}^{1/2} 6(x-x^2) dx}{\int_{1/3}^{2/3} 6(x-x^2) dx}$$

$$= \frac{[3x^2 - 2x^3]_{1/3}^{1/2}}{[3x^2 - 2x^3]_{1/3}^{2/3}}$$

$$= \frac{\left(\frac{3}{4} - \frac{2}{8}\right) - \left(\frac{3}{9} - \frac{2}{27}\right)}{\left(\frac{12}{9} - \frac{16}{27}\right) - \left(\frac{3}{9} - \frac{2}{27}\right)}$$

$$= \left[\frac{1}{2}\right] \text{ Ans}$$

(Q-4) Let X be a random variable with the expectation following probability distribution:-

x	-3	6	9
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find $E(X)$ and $E(X^2)$ and using the laws of expectation, evaluate $E(2X+1)^2$.

Sol-

$$E(X) = \sum_{i=1}^{\infty} x_i p(x_i)$$

$$= (-3)\left(\frac{1}{6}\right) + (6)\left(\frac{1}{2}\right) + (9)\left(\frac{1}{3}\right)$$

$$= -\frac{1}{2} + 3 + 3$$

$$E(X) = 5.5$$

$$\begin{aligned} \times E(X^2) &= \sum_{i=1}^{\infty} x_i^2 p(x_i) \\ &= (-3)^2 \left(\frac{1}{6}\right) + (6)^2 \left(\frac{1}{2}\right) + (9)^2 \left(\frac{1}{3}\right) \end{aligned}$$

$$E(X^2) = 46.5 \text{ Ans}$$

$$\begin{aligned} \times E(2X+1)^2 &= E(4X^2 + 1 + 4X) \\ &= 4E(X^2) + 1 + 4E(X) \\ &= 4(46.5) + 1 + 4(5.5) \end{aligned}$$

$$E(2X+1)^2 = 209 \text{ Ans}$$

Que. 5 If the Chebyshev's inequality for the random variable X is given by $P(-2 < X < 8) \geq \frac{21}{25}$, find $E(X)$ and $\text{Var}(X)$

Solⁿ

$$E(X) = \mu$$

$$V(X) = \sigma^2$$

$$\text{given } P(-2 < X < 8) \geq \frac{21}{25} \quad \text{--- (1)}$$

By Chebyshev's inequality $\rightarrow$

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2} \quad \text{--- (2)}$$

Comparing (1) and (2)

$$1 - \frac{1}{k^2} = \frac{21}{25}$$

$$\frac{9}{25} = \frac{1}{k^2}$$

$$k = \frac{5}{3}$$

Comparing (1) and (2) $\rightarrow$

$$\mu - k\sigma = -2 \quad \text{--- (3)}$$

$$\mu + k\sigma = 8 \quad \text{--- (4)}$$

$$\text{eq (3) + eq (4) } \Rightarrow$$

$$2\mu = 6$$

$$\boxed{\mu = 3}$$

Putting values of μ and k in eq (3) $\Rightarrow$

$$3 - \frac{5}{3}(\sigma) = -2$$

$$\boxed{\sigma = 3}$$

hence

$$E(X) = \mu = 3$$

$$V(X) = \sigma^2 = 9$$

Ans

Section C

Q-6 A Random variable X has the following probability function:

Values of X	0	1	2	3	4	5	6	7
$P(X)$	0	k	$2k$	$3k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(i) find k .

Sol $\Rightarrow \therefore \sum P(X) = 1$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$9k + 10k^2 = 1$$

$$10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - 9k - 1 = 0$$

$$10k(k+1) - 1(k+1) = 0$$

$$(k+1)(10k-1) = 0$$

$$k = -1, \frac{1}{10}$$

$\therefore k$ should be positive

for $P(X)$ to be $\in [0, 1]$

So $k = \frac{1}{10}$ Ans

(ii) $P(X < 6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$

$$= 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100}$$

$$= \frac{81}{100} \text{ Ans}$$

$$\therefore P(X \geq 6) = 1 - P(X < 6)$$

$$= 1 - \frac{81}{100}$$

$$= \frac{19}{100} \text{ Ans}$$

(iii) determine the distribution function of X

$$F(x_i) = \sum_{i=x_i \leq x} p(x_i)$$

$$F(x < 0) = \boxed{0}$$

$$F(x < 1) = p(0) = \boxed{0}$$

$$F(x < 2) = p(0) + p(1) = 0 + \frac{1}{10} = \boxed{\frac{1}{10}}$$

$$F(x < 3) = 0 + \frac{1}{10} + \frac{2}{10} = \boxed{\frac{3}{10}}$$

$$F(x < 4) = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} = \frac{5}{10} = \boxed{\frac{1}{2}}$$

$$F(x < 5) = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} = \frac{8}{10} = \boxed{\frac{4}{5}}$$

$$F(x < 6) = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} = \boxed{\frac{81}{100}}$$

$$F(x < 7) = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} + \frac{2}{100} = \boxed{\frac{83}{100}}$$

~~$F(x < 8)$~~

$$F(x > 7) = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} + \frac{2}{100} + \frac{17}{100} = \boxed{1}$$

$$F_x(x) = \begin{cases} 0, & x < 0 \\ 0, & 0 \leq x < 1 \\ \frac{1}{10}, & 1 \leq x < 2 \\ \frac{3}{10}, & 2 \leq x < 3 \\ \frac{1}{2}, & 3 \leq x < 4 \\ \frac{4}{5}, & 4 \leq x < 5 \\ \frac{81}{100}, & 5 \leq x < 6 \\ \frac{83}{100}, & 6 \leq x < 7 \\ 1, & x \geq 7 \end{cases}$$

(iv) if $P(X \leq a) > \frac{1}{2}$, find the minimum value of a
soln

$$\text{As } P(X \leq 4) = \frac{4}{5} > \frac{1}{2}$$

$$\text{So } \boxed{\text{min of } a = 4} \text{ Ans}$$

Q-7 A random number x is denoted at random between the values 0 and 1 so that its probability density function is $f(x) = kx^2(1-x^3)$, where k is a constant. Find the value of k .
Using this value of k , find its mean and variance.

soln

$$f(x) = kx^2(1-x^3)$$

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 kx^2(1-x^3) dx = 1$$

$$k \int_0^1 x^2 - x^5 dx = 1$$

$$k \left[\frac{x^3}{3} - \frac{x^6}{6} \right]_0^1 = 1$$

$$k \left[\frac{1}{3} - \frac{1}{6} \right] = 1$$

$$\boxed{k = 6} \text{ Ans}$$

$$E(x) = \int_0^1 x f(x) dx$$

$$= \int_0^1 x (6x^2(1-x^3)) dx$$

$$= 6 \int_0^1 (x^3 - x^6) dx$$

$$= 6 \left[\frac{x^4}{4} - \frac{x^7}{7} \right]_0^1$$

$$= 6 \left[\frac{1}{4} - \frac{1}{7} \right]$$

$$\boxed{E(x) = \frac{9}{14}} \text{ Ans}$$

$$V(X) = E(X^2) - (E(X))^2 \text{ --- (1)}$$

$$\begin{aligned} E(X^2) &= \int_0^1 x^2 f(x) dx \\ &= 6 \int_0^1 (x^4 - x^7) dx \\ &= 6 \left[\frac{x^5}{5} - \frac{x^8}{8} \right]_0^1 \\ &= 6 \left[\frac{1}{5} - \frac{1}{8} \right] \\ &= \frac{18}{40} = \frac{9}{20} \end{aligned}$$

$$V(X) = \frac{9}{20} - \left(\frac{9}{14}\right)^2$$

$$V(X) = \frac{9}{245} \quad \text{Ans}$$

Q-9 If X and Y are independent random variable with mean 2, 3 and variance 1, 2 respectively. Find the mean and variance of the random variable $Z = 2X - 5Y$.

Sol

$$\begin{aligned} E(X) &= 2, E(Y) = 3 \\ V(X) &= 1, V(Y) = 2 \\ Z &= 2X - 5Y \end{aligned}$$

$$\begin{aligned} E(Z) &= E(2X - 5Y) \\ &= E(2X) - E(5Y) \\ &= 2E(X) - 5E(Y) \\ &= 2(2) - 5(3) \\ E(Z) &= -11 \end{aligned}$$

$$\begin{aligned} V(Z) &= V(2X - 5Y) \\ &= V(2X) + V(-5Y) \\ &= (2^2) V(X) + (-5)^2 V(Y) \\ &= 4(1) + 25(2) \\ V(Z) &= 54 \end{aligned}$$

Q-10 A symmetric dice is thrown 600 times. Find the lower bound for the probability of getting 80 to 120 sixes.

Sol

$$\text{probability to get 1 six} = p = \frac{1}{6}$$

$$\text{given } n = 600$$

By Binomial Distribution $\Rightarrow$

$$E(X) = \mu = np = 600 \times \frac{1}{6} = \underline{\underline{100}}$$

$$V(X) = npq = np(1-p) \\ = 600 \times \frac{1}{6} \times \left(1 - \frac{1}{6}\right)$$

$$\sigma^2 = V(X) = \underline{\underline{\frac{250}{3}}}$$

$$\text{So } \boxed{\mu = 100}, \boxed{\sigma = \sqrt{\frac{250}{3}}}$$

$$\text{Now } P(80 < X < 120) = ?$$

using Chebyshev's inequality $\rightarrow$

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

Comparing $\rightarrow$

$$\mu - k\sigma = 80$$

$$100 - k\sqrt{\frac{250}{3}} = 80$$

$$k\sqrt{\frac{250}{3}} = -20$$

$$\boxed{k = -20\sqrt{\frac{3}{250}}}$$

Rejected $\times$
k can't be negative

$$\mu + k\sigma = 120$$

$$100 - k\sqrt{\frac{250}{3}} = 120$$

$$k\sqrt{\frac{250}{3}} = 20$$

$$k = 20\sqrt{\frac{3}{250}}$$

$$\boxed{k = 4\sqrt{\frac{3}{10}}}$$

hence

$$P(80 < X < 120) \geq 1 - \frac{1}{k^2}$$

$$\geq 1 - \frac{1}{\left(4\sqrt{\frac{3}{10}}\right)^2}$$

$$\geq 1 - \frac{1}{\frac{24}{5}}$$

$$P(80 < X < 120) \geq \underline{\underline{\frac{19}{24}}}$$

So lower bound for the probability of getting 80 to 120 sixes $= \frac{19}{24}$

Q-11 The distribution function of continuous random variable X given by

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{2}, & 0 \leq x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{x}{4}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

Find probability density function of X .

Sol $\rightarrow$ * $x < 0 \rightarrow$

$$f(x) = \frac{d}{dx} (F(x))$$

$$f(x) = \frac{d}{dx} (0) = 0$$

* $0 \leq x < 1 \rightarrow$

$$f(x) = \frac{d}{dx} \left(\frac{x}{2} \right) = \frac{1}{2}$$

* $1 \leq x < 2 \rightarrow$

$$f(x) = \frac{d}{dx} \left(\frac{1}{2} \right) = 0$$

* $2 \leq x < 4 \rightarrow$

$$f(x) = \frac{d}{dx} \left(\frac{x}{4} \right) = \frac{1}{4}$$

* $x \geq 4 \rightarrow$

$$f(x) = \frac{d}{dx} (1) = 0$$

Ans PDF $\Rightarrow$

$$f(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}, & 0 \leq x < 1 \\ 0, & 1 \leq x < 2 \\ \frac{1}{4}, & 2 \leq x < 4 \\ 0, & x \geq 4 \end{cases}$$

Ans

Q-12

Suppose that the life in hours of a certain part of radio tube is a continuous random variable X with pdf given by

$$f(x) = \begin{cases} \frac{100}{x^2}, & \text{when } x \geq 100 \\ 0, & \text{elsewhere} \end{cases}$$

- (i) What is probability that all of three such tubes in a given radio set will have to be replaced during the first 150 hours of operation?
- (ii) What is probability that none of three of the original tubes will have to be replaced during that first 150 hours of operation?
- (iii) What is probability that a tube will last less than 200 hours if it is known that the tube is still functioning after 150 hours of service?
- (iv) What is the maximum number of tubes that may be inserted into a set so that there is a probability of 0.5 that after 150 hours of service all of them are still functioning?

Sol: (i) probability to Replace one tube after 150 hour $\Rightarrow$

$$P(X \leq 150) = \int_{100}^{150} \frac{100}{x^2} dx$$

$$= \left[\frac{100}{x} \right]_{100}^{150}$$

$$= -\frac{100}{150} + \frac{100}{100}$$

$$= \frac{1}{3}$$

$$\text{probability to Replace 3 tube after 150 hour} = \left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = \boxed{\frac{1}{27}}$$

(ii) Probability to ^{not} replace one tube after 150 hour $\Rightarrow$

$$P(X > 150) = 1 - P(X \leq 150)$$

$$= 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

$$\text{Probability to replace none of three tube after 150 hour} = \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right) = \boxed{\frac{8}{27}}$$

$$(iii) P(X < 200 | X > 150) = \frac{P(150 < X < 200)}{P(X > 150)}$$

$$= \frac{\int_{150}^{200} \frac{100}{x^2} dx}{\frac{2}{3}}$$

$$= \frac{\left(-\frac{100}{200} + \frac{100}{150}\right)}{\frac{2}{3}}$$

$$= \frac{\frac{1}{6}}{\frac{2}{3}}$$

$$P(X < 200 | X > 150) = \frac{1}{4} \quad \text{Ans}$$

$$(iv) \text{ Probability of working one tube after 150 hour} = P(X > 150) = \frac{2}{3}$$

$$\text{probability of } n \text{ tube working after 150 hour} = \left(\frac{2}{3}\right)^n = 0.5$$

$$\left(\frac{2}{3}\right)^n = \frac{1}{2}$$

$$n \log \frac{2}{3} = \log \frac{1}{2}$$

$$n = \frac{\log \frac{1}{2}}{\log \frac{2}{3}}$$

$$n = 1.709$$

$$\text{max num of tube} = 1$$

Q-13 for geometric distribution $P(X) = 2^{-x}$, $x = 1, 2, 3, \dots$ prove that chebyshev's inequality gives $P\{|X-2| \leq 2\} > \frac{1}{2}$ while actual probability is $\frac{15}{16}$.

Let $\Rightarrow$ mean $= E(X) = \boxed{\mu = 2}$

$$V(X) = \sigma^2 = E(X^2) - (E(X))^2 \quad \text{--- (1)}$$

$$\begin{aligned} E(X^2) &= \sum_{x=1}^{\infty} x^2 \left(\frac{1}{2}\right)^x \\ &= \frac{\left(\frac{1}{2}\right) \left(1 + \frac{1}{2}\right)}{\left(\frac{1}{2}\right)^3} \\ &= \frac{\left(\frac{1}{2}\right) \left(\frac{3}{2}\right)}{\frac{1}{8}} \end{aligned}$$

$$\boxed{E(X^2) = 6}$$

using formula

$$\sum_{x=1}^{\infty} x^2 r^x = \frac{r(1+r)}{(1-r)^3}$$

So $V(X) = \sigma^2 = 6 - (2)^2$

$$\sigma^2 = 6 - 4$$

$$\sigma^2 = 2$$

$$\boxed{\sigma = \sqrt{2}}$$

Now to find (prove) $\rightarrow$

$$P(|X-2| \leq 2) > \frac{1}{2}$$

LHS $\Rightarrow$ $\Rightarrow P(-2 \leq X-2 \leq 2)$

$\Rightarrow P(0 \leq X \leq 4)$

By Chebyshev's inequality -

$$P(\mu - k\sigma \leq X \leq \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

Comparing LHS $\Rightarrow$

$$\mu - k\sigma = 0$$

$$2 - k\sqrt{2} = 0$$

$$\boxed{k = \sqrt{2}}$$

hence

$$P(0 \leq X \leq 4) \geq 1 - \frac{1}{(\sqrt{2})^2}$$

$$\geq 1 - \frac{1}{2}$$

$$\geq \frac{1}{2}$$

$$P(0 \leq X \leq 4) \geq \frac{1}{2} \Rightarrow \text{RHS}$$

Hence proved.

Now
actual
probability

$$P(0 \leq X \leq 4) = P(0) + P(1) + P(2) + P(3) + P(4) \\ = 0 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$$

Actual
probability

$$P(0 \leq X \leq 4) = \boxed{\frac{15}{16}} \text{ Ans}$$

Q-15

A man with n keys want to open his door and tries the keys independently and at random. Find the mean and variance of the numbers of trials required to open the door,

- (i) if unsuccessful keys are not eliminated from further selection and
(ii) if they are.

Soln

(i) Let X = no. of trials to open door

$$p = \frac{1}{n} \quad q = 1 - \frac{1}{n} = \frac{n-1}{n}$$

$$E(X) = \frac{1}{p} = \frac{1}{1/n} = n$$

$$\text{Var}(X) = \frac{1-p}{p^2} = \frac{1 - \frac{1}{n}}{1/n^2} = \frac{n-1}{1/n} = n^2 - n$$

⊗

(i) Let X be the numbers of trials.

$$\text{Then } P(X=1) = \frac{1}{n}$$

$$P(X=2) = \frac{n-1}{n} \cdot \frac{1}{n-1} = \frac{1}{n}$$

$$\vdots$$

$$P(X=n) = \frac{n-1}{n} \cdots \frac{1}{2} = \frac{1}{n}$$

The RV X a discrete uniform random variable in integer values $\{1, 2, \dots, n\}$

$$E(X) = \sum_{x=1}^n \frac{x}{n} = \frac{1}{n} \sum_{x=1}^n x = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \boxed{\frac{n+1}{2}}$$

$$E(X^2) = \sum_{x=1}^n \frac{x^2}{n} = \frac{1}{n} \frac{n(n+1)(2n+1)}{6}$$

$$\text{var}(X) = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

Q-16 Let $Y = X^2 + 2X$, where X is a random variable whose probability distribution is given as: x .

x	1	2	3	4
$P(X)$	0.1	0.2	0.5	0.1

Then find probability distribution, CDF, mean, and variance of Y .

Sol $\Rightarrow \because Y = X^2 + 2X$

$\times$ at $x=1$, $Y = (1)^2 + 2(1)$

$Y = 3$

$\times$ at $x=2$, $Y = (2)^2 + 2(2)$

$Y = 8$

$\times$ at $x=3$, $Y = (3)^2 + 2(3)$

$Y = 15$

$\times$ at $x=4$, $Y = (4)^2 + 2(4)$

$Y = 24$

Sol $\Rightarrow$

Probability distribution $\rightarrow$

Y	3	8	15	24
$P(Y)$	0.1	0.2	0.5	0.1

$\times$ CDF $\Rightarrow F_Y(y) \Rightarrow$

$F_Y(3) = P(Y \leq 3) = 0.1$

$F_Y(8) = P(Y \leq 8) = P(Y=3) + P(Y=8) = 0.1 + 0.2 = 0.3$

$F_Y(15) = P(Y \leq 15) = P(Y=3) + P(Y=8) + P(Y=15) = 0.1 + 0.2 + 0.5 = 0.8$

$F_Y(24) = P(Y \leq 24) = P(Y=3) + P(Y=8) + P(Y=15) + P(Y=24)$
 $= 0.1 + 0.2 + 0.5 + 0.1 = 0.9$

$F_Y(y) = \begin{cases} 0.1, & y \leq 3 \\ 0.3, & y \leq 8 \\ 0.8, & y \leq 15 \\ 0.9, & y \leq 24 \end{cases}$

$$\therefore \text{mean} = E(Y) = \sum y \cdot p(y)$$

$$E(Y) = (0.1 \times 3) + (0.2 \times 8) + (0.5 \times 15) + (0.1 \times 24)$$

$$E(Y) = 0.3 + 1.6 + 7.5 + 2.4 = \underline{\underline{11.8}}$$

$$\therefore \text{variance} = V(Y) = E(Y^2) - (E(Y))^2$$

$$V(Y) = (0.1 \times 9) + (0.2 \times 64) + (0.5 \times 225) + (0.1 \times 576) - (11.8)^2$$

$$V(Y) = \underline{\underline{44.56}}$$

Q-8

If t is any positive real number, show that the function defined by $p(x) = e^{-t} (1 - e^{-t})^{x-1}$ can represent a probability function of a random variable X assuming the values $1, 2, 3, \dots$ find the $E(X)$ and $\text{var}(X)$ of the distribution.

Sol $\Rightarrow$

Conditions for PMF $\Rightarrow$

$$\ast p(x) \geq 0 \text{ for all } x = 1, 2, 3, \dots \checkmark$$

$$\Rightarrow p(x) = \underbrace{e^{-t}}_{\substack{\text{always} \\ +ve \text{ for} \\ x \geq 1}} \underbrace{(1 - e^{-t})^{x-1}}_{\substack{\text{always} \\ +ve \text{ for} \\ x \geq 1}}$$

$$\ast \sum_{x=1}^{\infty} p(x) = 1$$

$$\Rightarrow \sum_{x=1}^{\infty} p(x) = \sum_{x=1}^{\infty} e^{-t} (1 - e^{-t})^{x-1}$$

$$= e^{-t} \sum_{x=1}^{\infty} (1 - e^{-t})^{x-1}$$

gp with $a=1$
 $r = (1 - e^{-t})$

$$= e^{-t} \frac{1}{1 - (1 - e^{-t})}$$

$$= \frac{e^{-t}}{e^{-t}}$$

$$= 1 \checkmark$$

$\Rightarrow$ hence, $p(x)$ can represent a probability function. Ans

$$* E(X) = \sum_{x=1}^{\infty} x p(x)$$

$$= \sum_{x=1}^{\infty} x e^{-t} (1-e^{-t})^{x-1}$$

$$= e^{-t} \sum_{x=1}^{\infty} x (1-e^{-t})^{x-1}$$

$$= e^{-t} \cdot \frac{1}{(e^{-t})^2}$$

$$\left[\because \sum_{x=1}^{\infty} x r^{x-1} = \frac{1}{(1-r)^2} \right]$$

for $|r| < 1$

$$\boxed{E(X) = e^t}$$

$$* E(X^2) = \sum_{x=1}^{\infty} x^2 p(x)$$

$$= \sum_{x=1}^{\infty} x^2 e^{-t} (1-e^{-t})^{x-1}$$

$$= e^{-t} \sum_{x=1}^{\infty} x^2 (1-e^{-t})^{x-1}$$

$$= e^{-t} \frac{1 + (1-e^{-t})}{(1 - (1-e^{-t}))^3}$$

$$\left[\because \sum_{x=1}^{\infty} x^2 r^{x-1} = \frac{1+r}{(1-r)^3} \right]$$

for $|r| < 1$

$$\boxed{E(X^2) = \frac{2-e^{-t}}{e^{-2t}}}$$

$$\begin{aligned}
 * \text{Var}(X) &= E(X^2) - (E(X))^2 \\
 &= \frac{(2 - e^{-t})}{e^{-2t}} - \left(\frac{1}{e^{-t}}\right)^2
 \end{aligned}$$

$$\boxed{\text{Var}(X) = \frac{1 - e^{-t}}{(e^{-t})^2}} \quad \text{Ans}$$